

Lecture 5: Models Do What They're Told

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5.1 Case Study: Time Through the Order Penalty

In Game 6 of the 2020 World Series, the Tampa Bay Rays' manager, Kevin Cash, pulled his starting pitcher, Blake Snell, midway through the sixth inning. When he was pulled, Snell had been pitching extremely well; he had allowed just two hits and struck out nine batters on 73 pitches. Moreover, the Rays had a one run lead. Snell's replacement, Nick Anderson, promptly gave up two runs, which ultimately proved decisive: the Rays went on to lose the game and the World Series. After the game, Cash justified his decision to pull Snell, remarking that he "didn't want Mookie [Betts] or [Corey] Seager seeing Blake a third time" (Rivera, 2020).

In his justification, Cash cites the *Time Through the Order Penalty* (TTOP), which is formally identified in Tango et al. (2007, pp. 187-190) and recently popularized by Lichtman (2013). It has long been observed that, on average, batters tend to perform better the more times they face a pitcher; for instance, they tend to get on base more often on their third time facing a pitcher than their second. Tango et al. (2007) quantified the corresponding drop-off in pitcher performance as increases in *weighted on-base average* (wOBA). They observed that the average wOBA of a plate appearance in the first time through the order (1TTO) is about 9 wOBA points less than in the second TTO (2TTO). Further, the average wOBA of a plate appearance in the second TTO is about 8 wOBA points less than that in the third TTO (3TTO) (Tango et al. 2007, Table 81).

wOBA overcomes many limitations of traditional metrics like batting average, on-base percentage, and slugging percentage. Briefly, batting average and on-base percentage treat all hits equally, with singles being worth as much as triples. Slugging percentage attempts to reward different types of hits differently, but does so in too simplistic of a fashion: in computing slugging percentage, a triple is worth three times what a single is worth. Such weighting is arbitrary, and is not tied to the relative impact of a triple over a single with regard to, say, run scoring or win probability. wOBA combines the different aspects of offensive production into one metric, weighing each offensive action in proportion to its actual run value (Slowinski, 2010). The wOBA of a plate appearance is simply the weight associated with the offensive action of the outcome. Specifically, the 2019 wOBA weight of each offensive action in decreasing order is 1.940 for a home run (HR), 1.529 for a triple (3B), 1.217 for a double (2B), 0.870 for a single (1B), 0.719 for hit-by-pitch (HBP), 0.690 for unintentional walks (uBB), and 0 for an out (OUT) (Fangraphs, 2021). wOBA is rescaled so that the league average wOBA equals the league average on-base percentage. Throughout this lecture, we use 2019 wOBA weights for each season. Additionally, we usually refer to wOBA points, which is wOBA multiplied by 1000, to be consistent with the baseball community's use of wOBA.

The TTOP is considered canon by much of the baseball community. Announcers routinely mention the third time through the order during broadcasts and several managers regularly use that benchmark to justify their decisions to pull starting pitchers at the start of the third TTO. For instance, A.J. Hinch, who managed the Houston Astros from 2015 to 2019, noted "the third time through is very difficult for a certain caliber of pitchers to get through." Brad Ausmus, who managed the Detroit Tigers from 2014 to 2017, explained "the more times a hitter sees a pitcher, the more success that hitter is going to have" (Laurila, 2015).

Let's dig into the data ourselves to see if we can corroborate this claim. More specifically, we want to answer the following question: is the apparent third-time-through penalty really a discontinuity at batter 19, or is

it mostly a smooth deterioration over the course of the game?

5.2 Modeling the Time Through the Order Penalty

5.2.1 Data

We have a dataset containing every plate appearance from 2018-2019 against a starting pitcher in the first three times through the order. In total, we have 216,025 plate appearances. We have the following variables:

- `i`: The i^{th} plate appearance.
- `y_i`: wOBA of the i^{th} plate appearance (using 2019 wOBA weights).
- `t_i`: Batter sequence number ($1, 2, \dots, 27$).

Note that the first time through the order corresponds to $t_i \in \{1, 2, \dots, 9\}$, the second time through the order corresponds to $t_i \in \{10, 11, \dots, 18\}$, and the third time through the order corresponds to $t_i \in \{19, 20, \dots, 27\}$.

Before fitting anything, it is useful to be explicit about what each model is being told to look for.

Model	What it is told to look for	What it cannot capture well
TTO step model	Jumps at batters 10 and 19	Smooth within-game decline
Linear sequence model	A steady change across the 27 batters	Sharp TTO jumps
Combined model	A steady change plus TTO jumps	Complex nonlinear decline or pitcher effects

Table 5.1: Model specification determines which patterns the model can discover.

5.2.2 Exploring the Data

Let's start by looking at the average wOBA in each time through the order. We do this in R by binning the data by time through the order and then computing the average wOBA in each bin. We observe in Table 5.2 that a starting pitcher performs worse on average in 2TTO than in 1TTO, and worse on average in 3TTO than in 2TTO. This is consistent with the time through the order narrative.

Time Through Order	wOBA
First	.303
Second	.318
Third	.333

Table 5.2: Average wOBA by time through the order. We can see that batters perform progressively better each time they face a pitcher.

At this point, the data are consistent with a step-function story, but they do not yet tell us whether the apparent penalty is a sharp discontinuity at the TTO boundaries or a smoother within-game trend.

To see that more clearly, it helps to plot the observed mean wOBA at each batter sequence number. Figure 5.1 shows those 27 observed means. There is a clear upward trend overall, but it does not look like a pair of perfectly sharp jumps.

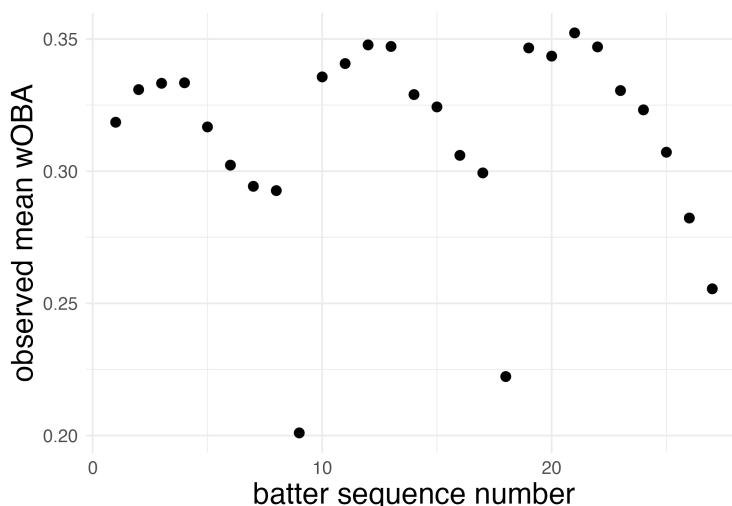


Figure 5.1: Observed mean wOBA at each batter sequence number.

5.2.3 Fitting the Model

Binning and averaging the data is equivalent to fitting the following regression model:

$$\mathbb{E}[y_i | t_i] = \beta_1 \mathbb{1}\{t_i \in 1TTO\} + \beta_2 \mathbb{1}\{t_i \in 2TTO\} + \beta_3 \mathbb{1}\{t_i \in 3TTO\}$$

where $\mathbb{1}\{z\}$ is the indicator function that is 1 if z is true and 0 otherwise.

Math HW: Prove this via $\hat{\beta} = (X^T X)^{-1} X^T y$.

We can verify this by fitting the model in R and checking the coefficients.

```
# fit the model
model = lm(formula = woba_19 ~ factor(TTO) + 0, data = mlb_data)
# get model summary
summary(model)
```

We can also re-parametrize the model slightly so β_2 represents pitcher decline from 1TTO to 2TTO, and β_3 represents pitcher decline from 2TTO to 3TTO. This is equivalent to fitting the following model:

$$\mathbb{E}[y_i | t_i] = \beta_1 + \beta_2 \mathbb{1}\{t_i \geq 2TTO\} + \beta_3 \mathbb{1}\{t_i \geq 3TTO\}$$

What about the [trajectory](#) across the batter sequence number $t = 1, 2, \dots, 27$? If the TTOP were truly driven by discrete batter-learning jumps, we would expect visible discontinuities around batter sequence numbers 9/10 and 18/19. Figure 5.2 overlays the fitted 3-level TTO model on top of the observed sequence-by-sequence means. Note this is equivalent to fitting the following model:

$$\mathbb{E}[y_i | t_i] = \sum_{t=1}^{27} \beta_t \mathbb{1}\{t_i = t\}$$

What's going on here? Why does the trajectory not fit along the fitted model? Lecture 2 gave us the right language for this: the columns of $\mathbf{X}$ determine what the model can represent. Right now, the model only has 3 step-function columns for the three TTO buckets, so it is forced to attribute everything to those three

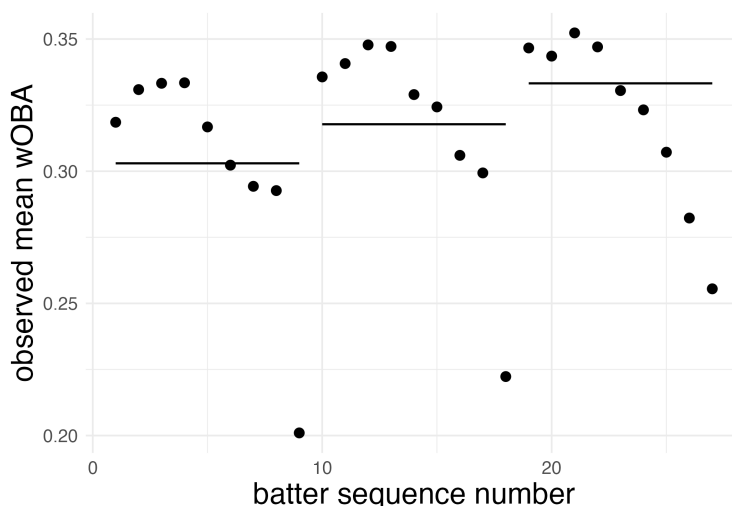


Figure 5.2: Observed mean wOBA by batter sequence number, with the fitted 3-level TTO step model overlaid.

levels. If other predictors also help explain wOBA, then leaving them out makes the TTO terms absorb too much structure. So before deciding whether the TTOP looks step-like or smooth, we should expand the design matrix.

5.3 Adding More Predictors

We now add a few more predictors that plausibly help forecast the outcome of a plate appearance:

- BQ_i : Quality of the i^{th} batter.
- PQ_i : Quality of the i^{th} pitcher.
- $hand_i$: Handedness match of the i^{th} pitcher-batter pair.
- $home_i$: Home field advantage.

For now, define a batter's quality by his end-of-season wOBA averaged across all his plate appearances, and likewise for pitchers. The goal is not to make a causal statement; the goal is to build a better predictive mean model by giving the regression more relevant columns.

Later in the course we will learn a better way to estimate batter and pitcher quality that doesn't use future data: **Empirical Bayes**.

5.3.1 Re-fitting the Model

We now specify a model that keeps the same 3 TTO step terms, but also includes these additional predictors. The model is given by:

$$\begin{aligned} \mathbb{E}[y_i | t_i] = & \beta_1 + \beta_2 \mathbb{1}\{t_i \geq 2TTO\} + \beta_3 \mathbb{1}\{t_i \geq 3TTO\} \\ & + \beta_{BQ} \cdot BQ_i + \beta_{PQ} \cdot PQ_i + \beta_{hand} \cdot hand_i + \beta_{home} \cdot home_i \end{aligned}$$

We proceed by fitting the model in R as follows:

```
# refit model with additional predictors
new_model = lm(formula = woba_19 ~ 1 + as.numeric(TTO >= 2) + as.numeric(TTO >= 3)
               + full_batter_woba_19 + full_pitcher_woba_19
               + hand_match + batter_home, data = mlb_data)

# get model summary
summary(new_model)
```

This richer step model still estimates that pitchers decline from one time through the order to the next. Tom Tango's original analysis is based off of this model and is a big part of the reason that starting pitchers are often removed in the 6th or 7th inning at the start of the third time through the order. Rather than plotting this 3-level fit by itself, we will carry it forward as a reference line when we compare richer trajectory models below.

5.3.2 Investigating Trajectory of Pitcher Decline

Once these extra predictors are in the model, what does the [trajectory](#) of pitcher decline look like across t ? We now revisit a lesson from Lectures 2 and 4: different basis terms tell the model to look for different shapes.

Indicator Model:

$$\mathbb{E}[y_i | t_i] = \sum_{t=1}^{27} \beta_t \mathbb{1}\{t_i = t\} + \beta_{BQ} \cdot BQ_i + \beta_{PQ} \cdot PQ_i + \beta_{hand} \cdot hand_i + \beta_{home} \cdot home_i$$

Linear Model:

$$\mathbb{E}[y_i | t_i] = \beta_0 + \beta_1 \cdot t_i + \beta_{BQ} \cdot BQ_i + \beta_{PQ} \cdot PQ_i + \beta_{hand} \cdot hand_i + \beta_{home} \cdot home_i$$

Additionally, we fit a smoothing spline over $\beta_1, \dots, \beta_{27}$. After fitting these models in R, we plot them together in Figure 5.3. Note that pitchers **do** decline on average from one time through the order to the next as Tango showed (gray step lines), but these richer models also reveal that pitchers on average [decline continuously](#) throughout the game. The indicator model is told to look for local changes at each batter sequence number, while the linear model is told to look for a smooth trend. The spline gives an even more flexible smooth curve. All 3 point in roughly the same direction.

These results suggest that a mechanical 3TTO cutoff may not be the right decision rule.

5.3.3 Takeaways

The main point here is about model specification, not causation. We changed the columns in $\mathbf{X}$ and changed the shape terms the model was allowed to use. Once we gave the regression both the extra predictors and smoother basis terms, the fitted pattern looked much more continuous than purely step-like.

A potential cause of continuous pitcher decline is pitcher fatigue, and a potential cause of discontinuous pitcher decline is batter learning. Is it possible that both are at play? We address this question in the next section.

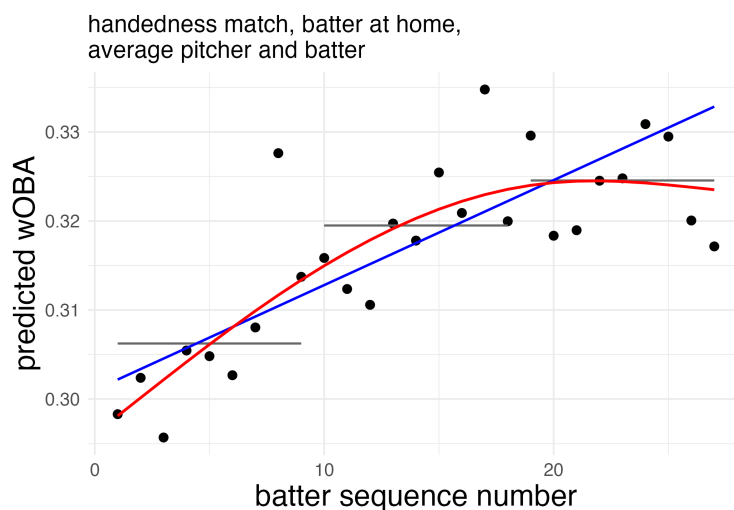


Figure 5.3: Fitted models of wOBA by batter sequence number after adding the extra predictors. The black dots are β_t from the indicator model, the gray horizontal lines are the fitted 3-level TTO step model, the blue line is the $\beta_0 + \beta_1 t$ from the linear model, and the red curve is the smoothing spline fit.

5.4 Continuous vs Discontinuous Pitcher Decline

5.4.1 Specifying the Combined Model

Now we combine the two ideas in one regression: a smooth within-game trend plus TTO step terms. In Lecture 2 language, we are simply adding both kinds of columns to the same design matrix. The model is:

$$\mathbb{E}[y_i | t_i] = \beta_0 + \beta_1 \cdot t_i + \beta_2 \cdot \mathbb{1}\{t_i \geq 2TTO\} + \beta_3 \cdot \mathbb{1}\{t_i \geq 3TTO\} \\ + \beta_{BQ} \cdot BQ_i + \beta_{PQ} \cdot PQ_i + \beta_{hand} \cdot hand_i + \beta_{home} \cdot home_i$$

Note that the **blue terms** represent continuous pitcher decline, the **red terms** represent discontinuous pitcher decline, and the black terms are the additional predictors we added earlier.

We fit the model in R as follows:

```
# fit combined model with both trend and step terms
combined_model = lm(formula = woba_19 ~ 1 + batter_sequence_number
                     + as.numeric(TTO >= 2) + as.numeric(TTO >= 3)
                     + full_batter_woba_19 + full_pitcher_woba_19
                     + hand_match + batter_home, data = mlb_data)

# get model summary
summary(combined_model)
```

When we look at the model summary, neither β_2 nor β_3 are statistically-significant. As a result, we do not find statistically-significant evidence of discontinuous pitcher decline between times through the order. Figure 5.4 illustrates this, where it's clear that continuous pitcher decline dominates any discontinuous decline.

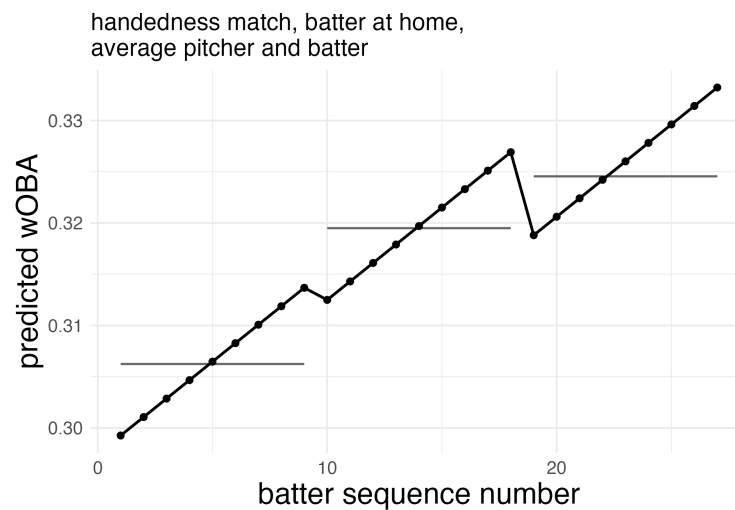


Figure 5.4: Predicted wOBA from the combined model after adding the extra predictors. The gray horizontal lines are the fitted 3-level TTO step model, and the black line adds a continuous within-game trend on top of it.

5.4.2 Takeaways

Models do what they're told. If you only tell the model to look for discontinuous pitcher decline, then that is what you'll find. When we look a bit deeper, we find that the expected wOBA forecast by our model increases steadily over the course of a game and does not display sharp discontinuities between times through the order. These results suggest that the conventional 3TTO cutoff should not be treated as a mechanical rule. A better decision rule would compare the current starter's projected performance against available relievers, conditional on game state, pitch count, bullpen quality, and personnel.

References

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